

Owen Chen

Acton, MA | 978-723-9696 | owenchen0408@gmail.com | owenchen.org

EDUCATION

Purdue University

B.S. Computer Engineering
GPA: 3.67/4.0; Dean's List & Semester Honors
SCALE Scalable Asymmetric Lifecycle Engagement

West Lafayette, IN

08/2025- 05/2028

EXPERIENCE

Birck Nanotechnology Center, Purdue University

Undergraduate Research Fellow - SCALE

West Lafayette, IN

05/2026 - Present

- Develop standardized measurement procedures and custom fixtures for UV-Vis-NIR diffuse-reflectance characterization of ESP32-based development boards obtained from authorized and third-party suppliers.
- Characterize 30 devices from 5 suppliers using VK-X3000 laser profilometry, extracting ISO 25178 surface parameters for supplier-level statistical comparison.
- Build Python pipelines to preprocess one-dimensional gas chromatograms and apply principal component analysis to evaluate supplier-level variation.

Research Fellow

01/2026 - 05/2026

- Developed and executed a High Temperature Gate Bias (HTGB) accelerated-aging protocol with AEC-Q101-aligned standards for onsemi power MOSFETs using Keithley source-measure units, Fourier-transform infrared spectroscopy, White Light Microscopy, and Keyence VK-X3000 laser profilometry.

Serelix Robotics

Engineering Intern

Bedford, MA

06/2025 - 09/2025

- Designed a parametric FDM-printed TPU foot interface for a post-stroke rehabilitation exoskeleton, using region-specific lattice geometries to tune stiffness, flexibility, and support.
- Designed a 9-channel ADS124S08 pressure-sensing PCB in KiCad.

Taipei Tech x MIT City Science Lab

UROP Researcher

Taipei, Taiwan

07/2024 - 01/2025

- Designed and built a prototype and a full-scale Wheeled-Bipedal Robot with a docking mechanism that reconfigured the system into a quadruped for improved static stability. <https://www.cs.taipeitech.com/en/project/blog-post-title-three-j5323>
- Implemented inverse kinematics for robot height and pose control and simulated linear-quadratic regulator and model predictive control strategies for balance and disturbance rejection on MuJoCo.
- Designed a two-layer power and CAN-bus distribution PCB in KiCad, sizing trace widths for expected load currents and reducing point-to-point wiring by 25%.

PROJECT AND COMPETITIONS

Purdue Semiconductor VIP at Birck

Junior Fabrication Technician

West Lafayette, IN

08/2026 - Present

- Develop and optimize chemical mechanical planarization (CMP) processes for copper through-silicon via (TSV) fabrication under Advanced Packaging & Heterogeneous Integration (AP/HI), Cleanliness and Integration Sub team.
- Integrate CMP with lithography, deposition, etching, and Cu electroplating to achieve planar wafer surfaces and establish a repeatable end-to-end TSV fabrication SOP.
- Operate the Park NX20 atomic force microscope (AFM) for nanoscale surface characterization and perform TAI (toluene, acetone, isopropanol) cleaning to prepare wafer for semiconductor fabrication and analysis.

FIRST Tech Challenge Team 16021 Techno Maniacs

Drone and Specialized Scoring Lead

Acton, MA

07/2022 - 09/2025

- Led the mechanical and controls integration of a drone-scoring subsystem, increasing maximum scoring capacity by approximately 20% and implementing Java-based actuator controls.
- Programmed and operated a WorkBee CNC router using Autodesk Fusion 360 CAM to machine aluminum and polymer drivetrain and scoring components.
- Awards: 3rd Place Inspire Award, 2025 New England Invitational; Winning Alliance, 2024 New England Invitational; Finalist Alliance, 2024 Massachusetts State Championship.

CERTIFICATIONS

• Google AI Fundamentals, Google, Jul. 2026 • PCB Basic Design Course, Altium Education, Mar. 2026 • Responsible Conduct of Research Training, CITI Program, Jan. 2026-Jan. 2030

SKILLS

Python, C++, Java, System Verilog, MATLAB; NumPy, pandas, SciPy, statistical analysis, Git, GitHub, Google Colab; Excel, Microsoft Office; ROS 2, OpenCV, PID, LQR, MPC; Arduino, ESP32, STM32; CAN, RS-485, KiCad, LTspice, PCB schematic design, PCB layout, soldering, power electronics, instrument control, data acquisition, electronic testing and validation, oscilloscopes, digital multimeters, SolidWorks, Siemens NX, Fusion 360, Rhino 3D, Grasshopper, FEA, CAM, CNC machining, MIG welding, 3D printing, laser profilometry, UV-Vis-NIR spectroscopy, FTIR spectroscopy, metrology, cleanroom operations, SOP development. Languages: English and Mandarin — native proficiency

RELEVANT COURSEWORK

• Differential Equations • Electrical Engineering Fundamentals II • Multivariable Calculus • Introduction to Digital System Design • Advanced C Programming